

Q. Discuss the basic organization of a computer system. Explain the primary functions of the Input, Output, Memory, ALU, and Control Units.

📌 **Definition to Remember**

The **Basic Organization of a Computer** relies on five fundamental, interconnected units that facilitate the flow of data: Input Unit, Output Unit, Memory Unit, Arithmetic and Logic Unit (ALU), and Control Unit (CU). Together, they accept, store, process, and output data.

1. The Five Functional Units

Unit	Primary Function	Examples
1. Input Unit	Accepts human-readable data/instructions, converts them into binary, and sends them to Memory.	Keyboard, Mouse, Scanner
2. Output Unit	Retrieves binary data from Memory, translates it to human-readable format, and displays it.	Monitor, Printer, Speaker
3. Memory Unit	Stores data and instructions. Divided into Primary (RAM - fast, volatile) and Secondary (HDD/SSD - permanent).	RAM, SSD
4. Arithmetic & Logic Unit (ALU)	The execution engine. Performs math (add, sub) and logical (AND, OR, compare) operations.	Inside CPU
5. Control Unit (CU)	The central nervous system. Coordinates all other units. Fetches, decodes, and issues signals to execute instructions.	Inside CPU

2. The Central Processing Unit (CPU)

The CPU is the "brain" of the computer. It is physically made up of:

- * **Control Unit (CU)**
- * **Arithmetic and Logic Unit (ALU)**
- * **Registers** (small, extremely fast internal memory).

3. Basic Data Flow

1. User types data → **Input Unit** → translates to binary.
2. Data is stored in **Memory Unit** (RAM).
3. **Control Unit** fetches instructions and directs data to the **ALU**.
4. **ALU** processes the data and sends results back to Memory.
5. **Output Unit** retrieves results from Memory and displays them to the user.

★ **Must-Write Points (for 10 marks)**

1. A computer is organized into 5 units: **Input, Output, Memory, ALU, Control Unit**.
2. **Input Unit**: Converts external data to binary and feeds it to the system.

3. **Output Unit:** Converts binary processed data back to human-readable form.
4. **Memory Unit:** Stores data/programs (Primary = RAM; Secondary = Hard Drive).
5. **ALU (Arithmetic Logic Unit):** Performs actual math and logical processing.
6. **CU (Control Unit):** Manages/coordinates all units (Fetches, Decodes, executes).
7. The **CPU** is the combination of the ALU, CU, and Registers.

⚡ Quick Recall

Computer Org → Input (data in) → Memory (storage) → CU (manager) → ALU (processor) →
Output (data out) → CPU = CU + ALU + Registers